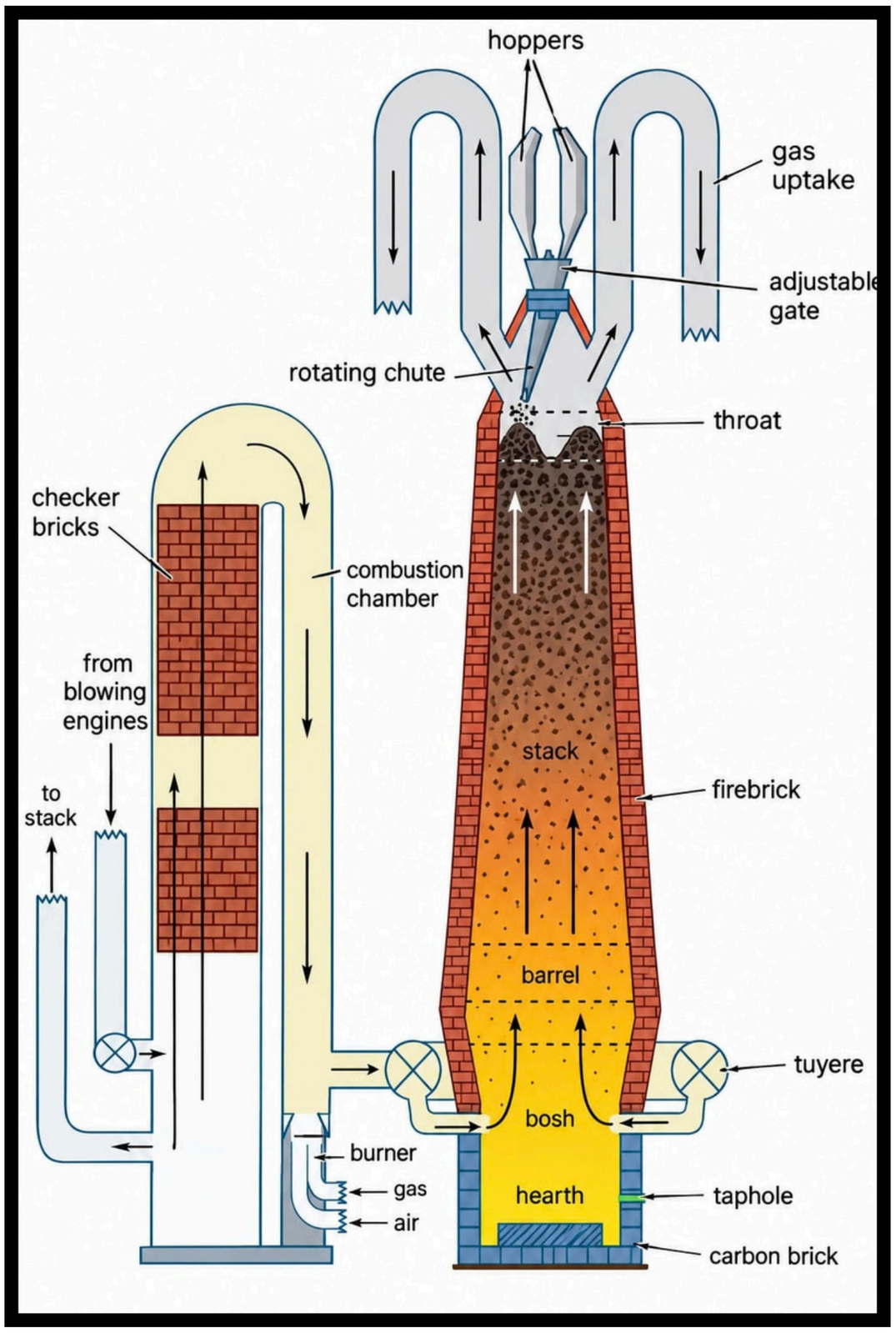
M & MT 2026
(Material & Manufacutring Technology International Conferance & Exibition)
Poster Presentation Competiton

Predicitve Maintenance in furnances



WHAT NEEDS MAINTENANCE?

- Refractory Lining: Wear and erosion
- Cooling Staves: Leakage and blockage
- Tuyeres: Damage and blockage
- Blowers & Motors: Bearing failure and overheating
- Pumps & Fans: Vibration and mechanical failure
- Gas System & Valves: Leakage and abnormal pressurre

KEY BENEFITS

- Reduced Downtime
- Improved Equipment Reliability
- Lower Maintenance Costs
- Enhanced Operational Safety
- Increased Production Efficiency

PREDICTIVE MAINTENANCE MONITORING

- Temperature Monitoring: Detect overheating and abnormal thermal patterns.
- Vibration Analysis: Identify bearing and rotating equipment faults.
- Pressure Monitoring: Detect abnormal pressure and possible leakage.
- Flow Monitoring: Identify cooling-water flow reduction.
- Current Monitoring: Detect motor overload and electrical abnormalities.
- Gas Analysis: Monitor CO, CO_2, H_2 and O_2 levels

KEY PARAMETERS TO MONITOR

- Temperature
- Vibration
- Pressure
- Water Flow
- Motor Current
- Gas Composition

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